Chapter 9: Wrapper Classes and Tokenizing Strings

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COSC 1437

Topic List Covered From Chapter 9

- Wrapper Classes (9.1 and 9.6)
  - What are they?
  - Why do we need them?
  - Autoboxing and Unboxing
- Tokenizing Strings (9.5)
  - String's split method
  - Delimiters

Wrapper Classes

Introduction

- Recall that Java has two types of variables: Primitives and References (Objects).
- Wrapper classes allow us to create **objects** that represent a primitive type.
- Java provides wrapper classes for all of the primitive types in `java.lang`, therefore no `import` statement is needed to use them. They are primarily the name of the primitive type, starting with a capital letter.
- You've already used these wrapper classes to parse data from an input dialog box (Integer.parseInt(), Double.parseDouble()).
- Wrapper objects are *immutable*, which means that once you create one, the value inside of it cannot change.

Primitive Data Type Wrappers

Wrapper Class	Primitive Type It Applies To
Byte	byte
Double	double
Float	float
Integer	int
Long	long
Short	short
Character	char

Creating a Wrapper Object

- To create objects from these wrapper classes, you can pass a value to the constructor:

```
Integer number = new Integer(7);
```

- You can also assign a primitive value to a wrapper class object:

```
Integer number;
number = 7;
```

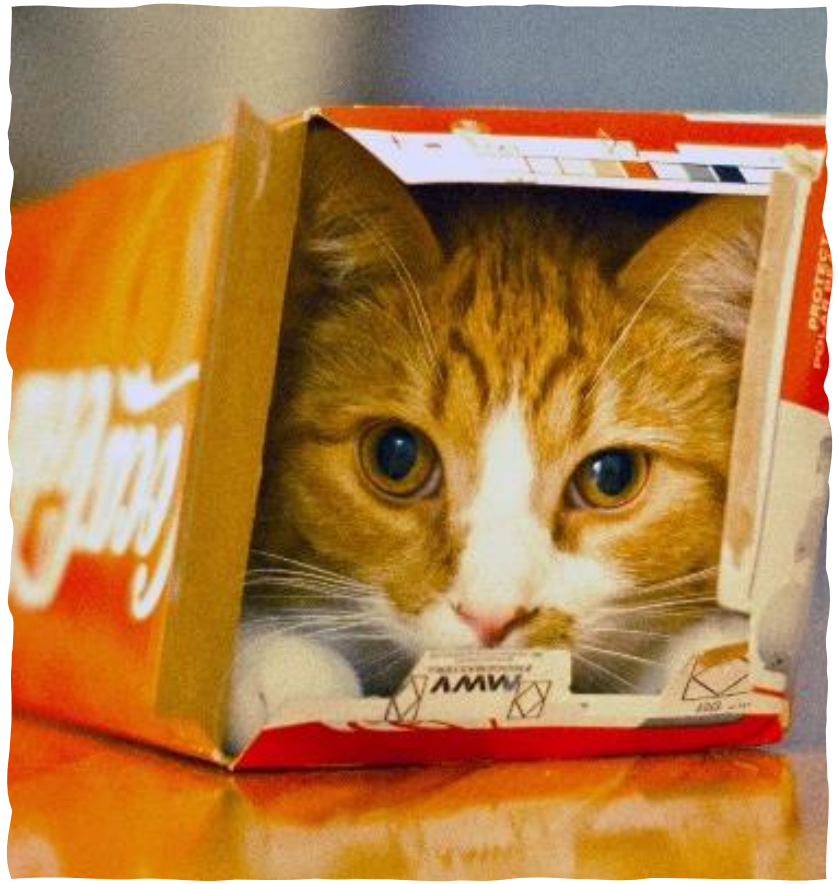
Wait a minute...

I just said this was ok:

```
Integer number;
number = 7;
```

But I'm assigning a primitive int value to an Integer object. Shouldn't that be an error?

Autoboxing



```
Integer number;
number = 7;
```

- You may think this is an error, but because `number` is a wrapper class object, *autoboxing* occurs.
- Autoboxing is Java's process of automatically "boxing up" a primitive value and putting it inside of a wrapper object

Autoboxing Examples

```
Double myNum = 7.8;
```

This creates a Double object named myNum that is a reference to a memory location on the heap with 7.8 inside of it. It is doing exactly the same thing as this line:

```
Double myNum = new Double(7.8);
```

```
Long myBigNumber;
myBigNumber = 123456789123456789;
```

This creates a Long object named myBigNumber on the first line, but it doesn't reference a value yet.

The second line creates the memory location on the heap and puts the value 123456789123456789 inside that memory location on the heap. It's doing the same thing as this line:

```
Long myBigNumber = new Long(123456789123456789);
```

Unboxing

- *Unboxing* does the opposite with wrapper class variables:

```
Integer myInt = 5;          // Autoboxes the value 5
int primitiveNumber;
primitiveNumber = myInt;    // unboxing
```

- Autoboxing takes a primitive and puts it **inside** a wrapper object
- Unboxing takes the value **out** of a wrapper object and puts it into a primitive variable

Unboxing at the North Pole

- You rarely need to declare numeric wrapper class objects, but they can be useful when you need to work with primitives in a context where primitives are not permitted.
- For example…remember `ArrayList` only work with objects.

```
ArrayList<int> list = new ArrayList<int>(); // Error!
ArrayList<Integer> list = new ArrayList<Integer>(); // OK!
```

- Wrapper classes also have many useful methods detailed throughout the chapter if you have time to read about them!

Tokenizing Strings

Concept

- Tokenizing a string is breaking a string down into its components, which are called *tokens*. The tokens in the string are separated by one or more *delimiters*.
- In the string, "Hamilton won every rap battle" there are:
- 5 tokens:
  - Hamilton
  - won
  - every
  - rap
  - battle
- And 1 delimiter:
  - The space character " "

String's split method

The String class has a method we can use to tokenize a string (break it down into its parts). It's called the split method, and works like this:

```
String str = "I'm looking for a mind at work!"; //Hamilton reference ☺

String[] tokens = str.split(" ");  //split the string on every space, and put each
                                   //token into a String array
```

To then print out all of the String tokens,

you can loop through the array:

```
for (String s : tokens)
    System.out.println(s);
```

Output:
I'm
looking
for
a
mind
at
work!

Delimiters

- A delimiter is a character that separates tokens.
- We saw that the space character is a typical delimiter, but you can use any character you like.
- Examples:
  - ```
    String nums = "3,7,8,13,4,6,11";   //This delimiter would be a comma
    String[] tokens = nums.split(",");  //split the string based on commas
    ```
  - ```
    String myDate = "04-01-2020";   //this delimiter would be a -
    String[] tokens = myDate.split("-");  //split the string based on dashes.
    ```

Using more than one delimiter

- You can split a string on more than one delimiter if you like:

```
String email = "Mandy.M.Mcdonald@somecompany.com";
```

- You can split this into all tokens where there are two delimiters:

  . and @

```
String[] tokens = email.split("[@.]");
```

- To use more than one delimiter, you enclose the ones you want inside of square brackets
  - If you're interested, this is called a *regular expression (regex)*

Practice

Write a small program to print out all of the tokens in this specific string:

```
/Users/profClark/Desktop/awesome.jpg
```

You'll need multiple delimiters, and you'll need a loop to print all the tokens.

Answer

```
public class StringSplit
{
    public static void main(String[] args)
    {
        String path = "/Users/profClark/Desktop/awesome.jpg";
        String[] tokens = path.split("[/.]");
        for (String s : tokens)
          System.out.println(s);
    }
}
```

You finished Chapter 9! Looks like you've made it onto the Nice List!